

Two-Qubit Gates in Superconducting Circuits

A First-Principles Derivation Reference

Quantum Computing with Superconducting Qubits — WMI / TUM

Based on Lecture 9 (Prof. Filipp, C. Schneider)

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1 Mathematical Foundations

1.1 The Rotating Frame Transformation

Given a Hamiltonian $H = H_0 + V$, we want a frame where H_0 disappears. Define the frame transformation as the **inverse** of free evolution:

$$U_0(t) = e^{+iH_0 t/\hbar} \quad (1)$$

The $+i$ sign is essential—it undoes the free precession $e^{-iH_0 t/\hbar}$.

The transformed state is $|\tilde{\psi}\rangle = U_0|\psi\rangle$. Starting from $i\hbar\partial_t|\psi\rangle = H|\psi\rangle$ and substituting $|\psi\rangle = U_0^\dagger|\tilde{\psi}\rangle$:

$$i\hbar\partial_t(U_0^\dagger|\tilde{\psi}\rangle) = (H_0 + V)U_0^\dagger|\tilde{\psi}\rangle \quad (2)$$

Expand the left side with the product rule:

$$i\hbar\dot{U}_0^\dagger|\tilde{\psi}\rangle + i\hbar U_0^\dagger\partial_t|\tilde{\psi}\rangle = (H_0 + V)U_0^\dagger|\tilde{\psi}\rangle \quad (3)$$

Since $U_0^\dagger = e^{-iH_0 t/\hbar}$, we have $i\hbar\dot{U}_0^\dagger = +H_0U_0^\dagger$. This cancels the H_0 on the right:

$$H_0U_0^\dagger|\tilde{\psi}\rangle + i\hbar U_0^\dagger\partial_t|\tilde{\psi}\rangle = H_0U_0^\dagger|\tilde{\psi}\rangle + VU_0^\dagger|\tilde{\psi}\rangle \quad (4)$$

Multiply both sides by U_0 from the left:

$$\boxed{i\hbar\partial_t|\tilde{\psi}\rangle = \underbrace{U_0 V U_0^\dagger}_{H_{\text{rot}}}|\tilde{\psi}\rangle} \quad (5)$$

The free part H_0 is gone. Only the interaction V , conjugated by U_0 , remains.

1.2 Conjugation of Pauli Operators

1.2.1 The corrected rotation table

For $U = e^{i\phi\sigma_z/2}$, the conjugation acts as an $\text{SO}(3)$ rotation on the Pauli vector. The correct formulas (verified by explicit 2×2 matrix multiplication) are:

Rotation around Z: $U = e^{i\phi\sigma_z/2}$

$$U \sigma_x U^\dagger = \cos \phi \sigma_x - \sin \phi \sigma_y \quad (6)$$

$$U \sigma_y U^\dagger = +\sin \phi \sigma_x + \cos \phi \sigma_y \quad (7)$$

$$U \sigma_z U^\dagger = \sigma_z \quad (8)$$

Rotation around X: $U = e^{i\phi\sigma_x/2}$

$$U \sigma_x U^\dagger = \sigma_x \quad (9)$$

$$U \sigma_y U^\dagger = \cos \phi \sigma_y - \sin \phi \sigma_z \quad (10)$$

$$U \sigma_z U^\dagger = +\sin \phi \sigma_y + \cos \phi \sigma_z \quad (11)$$

Proof for the Z-rotation of σ_x (to anchor the signs):

$$e^{i\phi\sigma_z/2} \sigma_x e^{-i\phi\sigma_z/2} = \begin{pmatrix} e^{i\phi/2} & 0 \\ 0 & e^{-i\phi/2} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{+i\phi/2} \end{pmatrix} = \begin{pmatrix} 0 & e^{+i\phi} \\ e^{-i\phi} & 0 \end{pmatrix} \quad (12)$$

Now $\cos \phi \sigma_x - \sin \phi \sigma_y = \begin{pmatrix} 0 & \cos \phi + i \sin \phi \\ \cos \phi - i \sin \phi & 0 \end{pmatrix} = \begin{pmatrix} 0 & e^{+i\phi} \\ e^{-i\phi} & 0 \end{pmatrix}$. Matches. ✓

1.2.2 The ladder operator identity

Since $\sigma_{\pm} = \frac{1}{2}(\sigma_x \pm i\sigma_y)$, applying the Z -rotation formulas:

$$\sigma_+ \rightarrow \frac{1}{2}[(\cos \phi \sigma_x - \sin \phi \sigma_y) + i(\sin \phi \sigma_x + \cos \phi \sigma_y)] = \frac{1}{2}[e^{i\phi} \sigma_x + ie^{i\phi} \sigma_y] = e^{i\phi} \sigma_+ \quad (13)$$

$$\boxed{e^{i\omega\sigma_z t/2} \sigma_{\pm} e^{-i\omega\sigma_z t/2} = e^{\pm i\omega t} \sigma_{\pm}} \quad (14)$$

The raising operator picks up $e^{+i\omega t}$, the lowering picks up $e^{-i\omega t}$.

1.2.3 Special case: X_{π} conjugation

The π -pulse around X is $X_{\pi} = e^{-i\pi\sigma_x/2} = -i\sigma_x$. Using the anticommutation $\{\sigma_x, \sigma_z\} = 0$:

$$X_{\pi} \sigma_z X_{\pi}^{\dagger} = \sigma_x \sigma_z \sigma_x = -\sigma_z \underbrace{\sigma_x \sigma_x}_I = -\sigma_z \quad (15)$$

Summary: X_{π} preserves σ_x , flips $\sigma_y \rightarrow -\sigma_y$ and $\sigma_z \rightarrow -\sigma_z$.

1.3 Tensor Product Rules

If $U = U_A \otimes U_B$ and $O = O_A \otimes O_B$:

$$U O U^{\dagger} = (U_A O_A U_A^{\dagger}) \otimes (U_B O_B U_B^{\dagger}) \quad (16)$$

Each qubit's conjugation is handled independently. This holds only when both U and O factorize as tensor products. If O is entangling (e.g. $|01\rangle\langle 10|$), we must work in the full space.

1.4 The Three-Level Frame Transformation

For a transmon with levels $|0\rangle, |1\rangle, |2\rangle$ and energies $E_0 = 0, E_1 = \hbar\omega_1, E_2 = \hbar(2\omega_1 + \Delta_1)$, the two-level σ_z is insufficient.

Why σ_z fails: Embedding σ_z in three dimensions gives $\sigma_z^{(3\text{-level})} = \text{diag}(1, -1, 0)$. Then $(\sigma_z^{(3\text{-level})})^2 = \text{diag}(1, 1, 0) \neq I_{3 \times 3}$, so the Pauli identity $e^{i\theta\sigma_z} = \cos \theta I + i \sin \theta \sigma_z$ is invalid. Direct computation shows $e^{i\phi\sigma_z^{(3\text{-level})}/2}$ gives $\text{diag}(e^{i\phi/2}, e^{-i\phi/2}, 1)$ —the $|2\rangle$ state gets phase 1 instead of $e^{i(2\omega_1 + \Delta_1)t}$, which is completely wrong.

The correct frame transformation is:

$$\boxed{U_1 = |0\rangle\langle 0| + e^{i\omega_1 t} |1\rangle\langle 1| + e^{i(2\omega_1 + \Delta_1)t} |2\rangle\langle 2|} \quad (17)$$

This assigns the **correct physical energy** to each level. Each ket-bra $|a\rangle\langle b|$ conjugated by this diagonal U picks up $e^{i(E_a - E_b)t/\hbar}$.

1.5 Block-Diagonal Exponentiation

If a Hamiltonian is block-diagonal: $H = H_1 \oplus H_2 \oplus \dots$, then:

$$e^{-iHt/\hbar} = e^{-iH_1 t/\hbar} \oplus e^{-iH_2 t/\hbar} \oplus \dots \quad (18)$$

For a 2×2 Pauli block $H = \hbar g \sigma_x$ (where $\sigma_x^2 = I$):

$$e^{-ig\sigma_x t} = \cos(gt) I - i \sin(gt) \sigma_x = \begin{pmatrix} \cos(gt) & -i \sin(gt) \\ -i \sin(gt) & \cos(gt) \end{pmatrix} \quad (19)$$

2 Gate Class 1: Flux-Controlled (Baseband) Gates

2.1 Starting Hamiltonian (Lab Frame)

Two capacitively coupled transmons, truncated to two levels:

$$\frac{H_{\text{lab}}}{\hbar} = \frac{\omega_A}{2} \sigma_z^{(A)} + \frac{\omega_B}{2} \sigma_z^{(B)} + J \sigma_x^{(A)} \sigma_x^{(B)} \quad (20)$$

Only two levels needed—this is a first-order resonant process.

2.2 Step 1: Expand $\sigma_x \sigma_x$

Write $\sigma_x = \sigma_+ + \sigma_-$ for each qubit:

$$\sigma_x^{(A)} \sigma_x^{(B)} = \underbrace{\sigma_+^{(A)} \sigma_-^{(B)} + \sigma_-^{(A)} \sigma_+^{(B)}}_{\text{swap (excitation-conserving)}} + \underbrace{\sigma_+^{(A)} \sigma_+^{(B)} + \sigma_-^{(A)} \sigma_-^{(B)}}_{\text{pair creation/annihilation}} \quad (21)$$

Four terms. All kept for now.

2.3 Step 2: Go to the Rotating Frame

Frame operator: Each qubit rotates at its own frequency:

$$U_0 = e^{i\omega_A \sigma_z^{(A)} t/2} \otimes e^{i\omega_B \sigma_z^{(B)} t/2} \quad (22)$$

Free Hamiltonians cancel: Since $U_0 = e^{iH_0 t/\hbar}$, the rotating-frame equation gives $H_{\text{rot}} = U_0 V U_0^\dagger$ with H_0 gone.

Transform each coupling term using $\sigma_\pm \rightarrow e^{\pm i\omega t} \sigma_\pm$ and the tensor product rule:

$$\sigma_+^{(A)} \sigma_-^{(B)} \rightarrow e^{+i\omega_A t} \sigma_+^{(A)} \cdot e^{-i\omega_B t} \sigma_-^{(B)} = e^{+i\delta t} \sigma_+^{(A)} \sigma_-^{(B)} \quad (23)$$

$$\sigma_-^{(A)} \sigma_+^{(B)} \rightarrow e^{-i\delta t} \sigma_-^{(A)} \sigma_+^{(B)} \quad (24)$$

$$\sigma_+^{(A)} \sigma_+^{(B)} \rightarrow e^{+i(\omega_A + \omega_B)t} \sigma_+^{(A)} \sigma_+^{(B)} \quad (25)$$

$$\sigma_-^{(A)} \sigma_-^{(B)} \rightarrow e^{-i(\omega_A + \omega_B)t} \sigma_-^{(A)} \sigma_-^{(B)} \quad (26)$$

where $\delta = \omega_A - \omega_B$.

2.4 Step 3: Rotating Wave Approximation (RWA)

The pair terms oscillate at $\omega_A + \omega_B \sim 10$ GHz, while $J \sim 10$ MHz. Over the gate timescale $\sim 1/J \sim 100$ ns, these terms oscillate ~ 1000 times and average to zero. Drop them:

$$\frac{H_{\text{rot}}}{\hbar} = J(e^{i\delta t} \sigma_+^{(A)} \sigma_-^{(B)} + e^{-i\delta t} \sigma_-^{(A)} \sigma_+^{(B)}) \quad (27)$$

2.5 Step 4: DC Flux Pulse ($\delta \rightarrow 0$)

The transmon frequency is $\omega_{01} \propto \sqrt{E_J(\Phi_{\text{ext}})}$ where $E_J(\Phi) = E_{J,\text{max}} |\cos(\pi\Phi/\Phi_0)|$. Applying a current pulse through Q1's SQUID flux line changes Φ_{ext} , tuning $\omega_A(\Phi) \rightarrow \omega_B$, making $\delta \rightarrow 0$.

At the **sweet spot** ($\Phi_{\text{ext}} = 0$), $\partial\omega/\partial\Phi = 0$ and the qubit is insensitive to flux noise. During the gate, the qubit leaves the sweet spot ($\partial\omega/\partial\Phi \neq 0$), reducing T_2 . This is the main fidelity bottleneck.

With $\delta = 0$, the exponentials become 1:

$$\frac{H_{\text{swap}}}{\hbar} = J(\sigma_+^{(A)} \sigma_-^{(B)} + \sigma_-^{(A)} \sigma_+^{(B)}) \quad (28)$$

2.6 Step 5: Matrix Representation and Block Structure

Using $\sigma_+ = |0\rangle\langle 1|$ and $\sigma_- = |1\rangle\langle 0|$ (where $|0\rangle$ is the higher-energy eigenstate of σ_z):

$$\sigma_+^{(A)}\sigma_-^{(B)} = |0\rangle\langle 1|_A \otimes |1\rangle\langle 0|_B = |01\rangle\langle 10| \quad (29)$$

$$\sigma_-^{(A)}\sigma_+^{(B)} = |1\rangle\langle 0|_A \otimes |0\rangle\langle 1|_B = |10\rangle\langle 01| \quad (30)$$

In the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$:

$$\frac{H_{\text{swap}}}{\hbar} = J \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (31)$$

The Hamiltonian conserves total excitation number. The block structure is:

$$H_{\text{swap}} = \underbrace{(0)}_{|00\rangle} \oplus \underbrace{\hbar J \sigma_x}_{\{|01\rangle, |10\rangle\}} \oplus \underbrace{(0)}_{|11\rangle} \quad (32)$$

$|00\rangle$ and $|11\rangle$ are annihilated (no excitation to swap, or both excited so no room to swap within two levels).

2.7 Step 6: Exponentiate

Exponentiate each block. The trivial blocks give 1. The 2×2 block gives $e^{-iJt\sigma_x}$:

$$U(t) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(Jt) & -i\sin(Jt) & 0 \\ 0 & -i\sin(Jt) & \cos(Jt) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (33)$$

2.8 Step 7: Special Gate Times

At $Jt = \pi/4$: $\sqrt{i\text{SWAP}}$, creating maximal entanglement from $|01\rangle$:

$$|01\rangle \rightarrow \frac{1}{\sqrt{2}}(|01\rangle - i|10\rangle) \quad (34)$$

At $Jt = \pi/2$: full iSWAP:

$$U_{\text{iSWAP}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (35)$$

The $-i$ (vs. $+i$ in some conventions) comes from $e^{-i\pi\sigma_x/2} = -i\sigma_x$; the sign depends on the coupling sign.

Why the i ? Unitary evolution $e^{-i\theta\sigma_x}$ always introduces phase factors— $\cos$ and $-i\sin$. At full swap, $\cos(\pi/2) = 0$ and $-i\sin(\pi/2) = -i$. For a single qubit this global $-i$ is unobservable, but for two qubits it creates entanglement: $|01\rangle \rightarrow -i|10\rangle$ has a relative phase with $|00\rangle$ that cannot be removed.

2.9 Step 8: Virtual-Z Calibration

The raw gate has parasitic single-qubit phases from Stark shifts during the flux pulse. These are corrected by virtual- Z rotations (see Section 7).

3 Gate Class 2: Cross-Resonance (Microwave-Activated)

3.1 Starting Hamiltonian (Lab Frame)

Two fixed-frequency transmons (Q1 = control, Q2 = target) with exchange coupling J (already RWA'd from $\sigma_x \sigma_x$, dropping pair terms at $\omega_1 + \omega_2$). Q1 is driven at Q2's frequency ω_2 :

$$\frac{H_{\text{lab}}}{\hbar} = \frac{\omega_1}{2} \sigma_z^{(1)} + \frac{\omega_2}{2} \sigma_z^{(2)} + J(\sigma_+^{(1)} \sigma_-^{(2)} + \text{h.c.}) + \Omega_1(t) \cos(\omega_2 t) \sigma_x^{(1)} \quad (36)$$

The third level $|2\rangle_1$ of Q1 is needed because this is a second-order virtual process where the anharmonicity plays an essential role.

3.2 Step 1: Choose the Rotating Frame (Frame B)

We rotate **both** qubits at the **drive frequency** $\omega_d = \omega_2$:

$$U_0 = e^{i\omega_2 \sigma_z^{(1)} t/2} \otimes e^{i\omega_2 \sigma_z^{(2)} t/2} \quad (37)$$

Why this frame? We compared two options:

Frame A (each qubit at own frequency): kills both free Hamiltonians, but makes both the coupling and the drive time-dependent at δ_{12} . Messy.

Frame B (both at ω_2): leaves a residual detuning $\frac{\delta_{12}}{2} \sigma_z^{(1)}$ on Q1 (since $\omega_1 - \omega_2 = \delta_{12} \neq 0$), but makes both the coupling and the drive time-independent. Clean.

Transform each term:

Q1 free: $\frac{\omega_1}{2} \sigma_z^{(1)} \rightarrow \frac{\omega_1 - \omega_2}{2} \sigma_z^{(1)} = \frac{\delta_{12}}{2} \sigma_z^{(1)}$ (residual detuning).

Q2 free: $\frac{\omega_2}{2} \sigma_z^{(2)} \rightarrow \frac{\omega_2 - \omega_2}{2} \sigma_z^{(2)} = 0$ (fully removed).

Exchange coupling: Both qubits rotate at the *same* frequency ω_2 , so the phases cancel:

$$\sigma_+^{(1)} \sigma_-^{(2)} \rightarrow e^{i(\omega_2 - \omega_2)t} \sigma_+^{(1)} \sigma_-^{(2)} = \sigma_+^{(1)} \sigma_-^{(2)} \quad (38)$$

The coupling is **unchanged and time-independent**.

Drive term: Use the Z -rotation formula: $U_1 \sigma_x^{(1)} U_1^\dagger = \cos(\omega_2 t) \sigma_x - \sin(\omega_2 t) \sigma_y$. Multiply by $\cos(\omega_2 t)$:

$$\Omega_1 \cos(\omega_2 t) [\cos(\omega_2 t) \sigma_x - \sin(\omega_2 t) \sigma_y] = \Omega_1 \left[\frac{1 + \cos(2\omega_2 t)}{2} \sigma_x - \frac{\sin(2\omega_2 t)}{2} \sigma_y \right] \quad (39)$$

RWA: Drop the $2\omega_2 \sim 10$ GHz terms (both σ_x and σ_y oscillating parts). Surviving:

$$\frac{\Omega_1}{2} \sigma_x^{(1)} \quad (40)$$

3.3 Step 2: The Rotating-Frame Hamiltonian

Rotating-Frame Hamiltonian (Frame B, after RWA)

$$\frac{H_{\text{rot}}}{\hbar} = \underbrace{\frac{\delta_{12}}{2} \sigma_z^{(1)}}_{H_0} + \underbrace{\frac{\Omega_1}{2} \sigma_x^{(1)}}_{H_d} + \underbrace{J(\sigma_+^{(1)} \sigma_-^{(2)} + \sigma_-^{(1)} \sigma_+^{(2)})}_{V_J} \quad (41)$$

Everything is time-independent. V_J is treated as a perturbation ($J \ll \delta_{12}$).

3.4 Step 3: The Schrieffer-Wolff Transformation

3.4.1 The idea

Find an anti-Hermitian generator S (with $S \sim J/\delta_{12} \ll 1$) such that:

$$[S, H_0] = -V_J \quad (42)$$

This eliminates the exchange coupling at first order. The transformed Hamiltonian is:

$$\tilde{H} = H_0 + H_d + \underbrace{[S, H_d]}_{\text{entangling (the prize)}} + \underbrace{\frac{1}{2}[S, V_J]}_{\text{freq. shifts + static ZZ}} + \underbrace{\mathcal{O}(J^2\Omega_1/\delta_{12}^2)}_{\text{dropped}} \quad (43)$$

The V_J and $[S, H_0]$ cancel by construction. Higher orders are suppressed by J/δ_{12} .

3.4.2 The two channels

Q1 has three levels, so V_J has two channels:

$$V_1 = \hbar J(|10\rangle\langle 01| + |01\rangle\langle 10|) \quad \text{Ch.1: } |01\rangle \leftrightarrow |10\rangle, \text{ gap } \delta_{12} \quad (44)$$

$$V_2 = \hbar\sqrt{2}J(|20\rangle\langle 11| + |11\rangle\langle 20|) \quad \text{Ch.2: } |11\rangle \leftrightarrow |20\rangle, \text{ gap } \delta_{12} + \Delta_1 \quad (45)$$

The $\sqrt{2}$ in Channel 2 is the bosonic factor from $a_1^\dagger|1\rangle_1 = \sqrt{2}|2\rangle_1$.

The generators satisfying $[S_i, H_0] = -V_i$ are:

$$S_1 = \frac{J}{\delta_{12}}(|10\rangle\langle 01| - |01\rangle\langle 10|) \quad (46)$$

$$S_2 = \frac{\sqrt{2}J}{\delta_{12} + \Delta_1}(|20\rangle\langle 11| - |11\rangle\langle 20|) \quad (47)$$

Each has the structure: coupling/denominator $\times$ (ket-bra $-$ h.c.), where the minus sign makes S anti-Hermitian.

3.4.3 The drive in two channels

The drive $\frac{\Omega_1}{2}\sigma_x^{(1)} \otimes I^{(2)}$ also splits because Q1 has three levels. Using $\sigma_x^{(3\text{-level})} = (|0\rangle\langle 1| + |1\rangle\langle 0|) + \sqrt{2}(|1\rangle\langle 2| + |2\rangle\langle 1|)$:

$$H_d^{01} = \frac{\hbar\Omega_1}{2}(|00\rangle\langle 10| + |10\rangle\langle 00| + |01\rangle\langle 11| + |11\rangle\langle 01|) \quad (48)$$

$$H_d^{12} = \frac{\sqrt{2}\hbar\Omega_1}{2}(|10\rangle\langle 20| + |20\rangle\langle 10| + |11\rangle\langle 21| + |21\rangle\langle 11|) \quad (49)$$

3.5 Step 4: Computing $[S, H_d]$ — The Entangling Interaction

Four commutators: $[S_1 + S_2, H_d^{01} + H_d^{12}]$. Two give zero in the computational subspace (they produce leakage to $|20\rangle$), two give the physics.

3.5.1 Commutator 1: $[S_1, H_d^{01}]$

Prefactor: $\frac{J}{\delta_{12}} \cdot \frac{\Omega_1}{2}$.

Compute $S_1 \cdot H_d^{01}$ using $|a\rangle\langle b| \cdot |c\rangle\langle d| = \delta_{bc}|a\rangle\langle d|$:

$$|10\rangle\langle 01| \cdot |01\rangle\langle 11| = |10\rangle\langle 11| \quad (\langle 01|01\rangle = 1) \quad (50)$$

$$-|01\rangle\langle 10| \cdot |10\rangle\langle 00| = -|01\rangle\langle 00| \quad (\langle 10|10\rangle = 1) \quad (51)$$

All other products have zero inner products. So $S_1 \cdot H_d^{01} = |10\rangle\langle 11| - |01\rangle\langle 00|$.

Compute $H_d^{01} \cdot S_1$:

$$|00\rangle\langle 10| \cdot |10\rangle\langle 01| = |00\rangle\langle 01| \quad (52)$$

$$|11\rangle\langle 01| \cdot (-|01\rangle\langle 10|) = -|11\rangle\langle 10| \quad (53)$$

So $H_d^{01} \cdot S_1 = |00\rangle\langle 01| - |11\rangle\langle 10|$.

The commutator:

$$[S_1, H_d^{01}] = |10\rangle\langle 11| + |11\rangle\langle 10| - |01\rangle\langle 00| - |00\rangle\langle 01| \quad (54)$$

Translate to Paulis: Using $|1\rangle\langle 1|_1 = \frac{I-Z}{2}$ and $|0\rangle\langle 0|_1 = \frac{I+Z}{2}$:

$$|10\rangle\langle 11| + |11\rangle\langle 10| = |1\rangle\langle 1|_1 \otimes \sigma_x^{(2)} = \frac{I-Z}{2} \otimes X \quad (55)$$

$$|00\rangle\langle 01| + |01\rangle\langle 00| = |0\rangle\langle 0|_1 \otimes \sigma_x^{(2)} = \frac{I+Z}{2} \otimes X \quad (56)$$

$$[S_1, H_d^{01}] = \frac{I-Z}{2} \otimes X - \frac{I+Z}{2} \otimes X = -ZX \quad (57)$$

With prefactor:

$$\boxed{[S_1, H_d^{01}] = -\frac{\hbar\Omega_1 J}{2\delta_{12}} ZX} \quad (58)$$

Pure ZX , coefficient J/δ_{12} . This is the only term at two levels.

3.5.2 Commutator 2: $[S_2, H_d^{12}]$

Prefactor: $\frac{\sqrt{2}J}{\delta_{12}+\Delta_1} \cdot \frac{\sqrt{2}\Omega_1}{2} = \frac{\Omega_1 J}{\delta_{12}+\Delta_1}$ (the two $\sqrt{2}$ factors multiply to 2).

Compute $S_2 \cdot H_d^{12}$:

$$|20\rangle\langle 11| \cdot |11\rangle\langle 21| = |20\rangle\langle 21| \quad (\text{leakage, drop}) \quad (59)$$

$$-|11\rangle\langle 20| \cdot |20\rangle\langle 10| = -|11\rangle\langle 10| \quad (\text{computational}) \quad (60)$$

Compute $H_d^{12} \cdot S_2$:

$$|10\rangle\langle 20| \cdot |20\rangle\langle 11| = |10\rangle\langle 11| \quad (\text{computational}) \quad (61)$$

$$|21\rangle\langle 11| \cdot (-|11\rangle\langle 20|) = -|21\rangle\langle 20| \quad (\text{leakage, drop}) \quad (62)$$

The commutator (computational subspace only):

$$[S_2, H_d^{12}]|_{\text{comp}} = -(|10\rangle\langle 11| + |11\rangle\langle 10|) = -\frac{I-Z}{2} \otimes X = -\frac{IX - ZX}{2} \quad (63)$$

With prefactor:

$$\boxed{[S_2, H_d^{12}]|_{\text{comp}} = -\frac{\hbar\Omega_1 J}{2(\delta_{12} + \Delta_1)}(IX - ZX)} \quad (64)$$

Channel 2 only acts when Q1 is in $|1\rangle$ (because $|2\rangle$ is accessible only from $|1\rangle$), producing $\frac{I-Z}{2} \otimes X$, which decomposes into both IX and ZX .

3.5.3 Cross terms: $[S_1, H_d^{12}]$ and $[S_2, H_d^{01}]$

S_1 acts on $\{|01\rangle, |10\rangle\}$ while H_d^{12} involves $|20\rangle, |21\rangle$. The only nonzero products are $|01\rangle\langle 20|$ and $|20\rangle\langle 01|$ —entirely outside the computational subspace:

$$[S_1, H_d^{12}]|_{\text{comp}} = 0, \quad [S_2, H_d^{01}]|_{\text{comp}} = 0 \quad (65)$$

3.6 Step 5: Extract μ and ν

Combine the two nonzero commutators:

$$[S, H_d]_{\text{comp}} = \underbrace{-\frac{\hbar\Omega_1 J}{2\delta_{12}} ZX}_{\text{from Ch.1}} + \underbrace{-\frac{\hbar\Omega_1 J}{2(\delta_{12} + \Delta_1)} IX + \frac{\hbar\Omega_1 J}{2(\delta_{12} + \Delta_1)} ZX}_{\text{from Ch.2}} \quad (66)$$

IX coefficient (only Channel 2 contributes):

$$-\frac{\hbar\Omega_1}{2} \cdot \frac{J}{\delta_{12} + \Delta_1} = -\frac{\hbar\Omega_1}{2} \cdot \frac{J}{\delta_{12}} \cdot \frac{\delta_{12}}{\delta_{12} + \Delta_1} \implies \boxed{\nu = \frac{J}{\delta_{12}} \cdot \frac{\delta_{12}}{\delta_{12} + \Delta_1}} \quad (67)$$

ZX coefficient (both channels, **opposite signs**—Channel 2 partially cancels Channel 1):

$$-\frac{\hbar\Omega_1 J}{2} \left[\frac{1}{\delta_{12}} - \frac{1}{\delta_{12} + \Delta_1} \right] = -\frac{\hbar\Omega_1}{2} \cdot \frac{J\Delta_1}{\delta_{12}(\delta_{12} + \Delta_1)} \implies \boxed{\mu = \frac{J}{\delta_{12}} \cdot \frac{\Delta_1}{\delta_{12} + \Delta_1}} \quad (68)$$

Sanity checks: $\mu + \nu = J/\delta_{12}$ (total coupling split between entangling and non-entangling). If $\Delta_1 \rightarrow 0$ (harmonic): $\mu \rightarrow 0$, no entangling gate—two virtual pathways cancel. If $|\Delta_1| \gg |\delta_{12}|$: $\mu \rightarrow J/\delta_{12}$, Channel 2 frozen out, full strength.

The CR Effective Hamiltonian

$$H_{\text{CR}} = \frac{\hbar\Omega_1}{2} [\alpha ZI - \nu IX - \mu ZX] \quad (69)$$

where α absorbs the detuning $\delta_{12}/2$ from H_0 , frequency shifts from $\frac{1}{2}[S, V_J]$, and Stark shifts from higher-order terms.

3.7 Step 6: Computing $\frac{1}{2}[S, V_J]$ — Frequency Shifts and Static ZZ

3.7.1 Cross terms vanish

$[S_1, V_2] = 0$ and $[S_2, V_1] = 0$ (no shared states between the two channels).

3.7.2 $\frac{1}{2}[S_1, V_1]$: two-level frequency shifts

Prefactor: $\frac{J}{\delta_{12}} \cdot \hbar J = \frac{\hbar J^2}{\delta_{12}}$.

Compute $S_1 \cdot V_1$:

$$|10\rangle\langle 01| \cdot |01\rangle\langle 10| = |10\rangle\langle 10| \quad (70)$$

$$-|01\rangle\langle 10| \cdot |10\rangle\langle 01| = -|01\rangle\langle 01| \quad (71)$$

$$S_1 \cdot V_1 = |10\rangle\langle 10| - |01\rangle\langle 01|.$$

Compute $V_1 \cdot S_1$: By the same logic: $V_1 \cdot S_1 = -|10\rangle\langle 10| + |01\rangle\langle 01|$.

Commutator: $[S_1, V_1] = 2|10\rangle\langle 10| - 2|01\rangle\langle 01|$.

Translate to Paulis:

$$|10\rangle\langle 10| = \frac{I-Z}{2} \otimes \frac{I+Z}{2} = \frac{1}{4}(II + IZ - ZI - ZZ) \quad (72)$$

$$|01\rangle\langle 01| = \frac{I+Z}{2} \otimes \frac{I-Z}{2} = \frac{1}{4}(II - IZ + ZI - ZZ) \quad (73)$$

$$|10\rangle\langle 10| - |01\rangle\langle 01| = \frac{1}{2}(IZ - ZI).$$

$$\boxed{\frac{1}{2}[S_1, V_1] = \frac{\hbar J^2}{2\delta_{12}}(IZ - ZI)} \quad (74)$$

Pure single-qubit frequency shifts. **No ZZ .**

3.7.3 $\frac{1}{2}[S_2, V_2]$: three-level correction (source of static ZZ)

Prefactor: $\frac{\sqrt{2}J}{\delta_{12}+\Delta_1} \cdot \hbar\sqrt{2}J = \frac{2\hbar J^2}{\delta_{12}+\Delta_1}$.

Compute $S_2 \cdot V_2$: $|20\rangle\langle 11| \cdot |11\rangle\langle 20| = |20\rangle\langle 20|$ and $-|11\rangle\langle 20| \cdot |20\rangle\langle 11| = -|11\rangle\langle 11|$.

Compute $V_2 \cdot S_2$: $V_2 \cdot S_2 = -|20\rangle\langle 20| + |11\rangle\langle 11|$.

Commutator: $[S_2, V_2] = 2(|20\rangle\langle 20| - |11\rangle\langle 11|)$ times the prefactor.

After combining: $\frac{1}{\hbar}[S_2, V_2] = \frac{4J^2}{\delta_{12}+\Delta_1}(|20\rangle\langle 20| - |11\rangle\langle 11|)$.

Project onto computational subspace (drop $|20\rangle\langle 20|$):

$$\frac{1}{2}[S_2, V_2]_{\text{comp}} = -\frac{2\hbar J^2}{\delta_{12} + \Delta_1}|11\rangle\langle 11| \quad (75)$$

Expand $|11\rangle\langle 11| = \frac{I-Z}{2} \otimes \frac{I-Z}{2} = \frac{1}{4}(II - IZ - ZI + ZZ)$:

$$\boxed{\frac{1}{2}[S_2, V_2]_{\text{comp}} = -\frac{\hbar J^2}{2(\delta_{12} + \Delta_1)}(II - IZ - ZI + ZZ)} \quad (76)$$

This contains ZZ—the static parasitic interaction. It comes **exclusively** from Channel 2 ($|11\rangle \leftrightarrow |20\rangle$), because this is the only virtual process that shifts $|11\rangle$'s energy relative to the other computational states.

3.7.4 Classification of all terms

Operator	Source	$\propto \Omega_1?$	Treatment
ZI	$H_0 + \frac{1}{2}[S, V_J]$	No (always on)	Virtual- Z on Q1
IZ	$\frac{1}{2}[S, V_J]$	No (always on)	Virtual- Z on Q2
ZZ	$\frac{1}{2}[S_2, V_2]$ only	No (always on)	Tunable coupler ($J_{\text{eff}} \rightarrow 0$)
IX	$[S_2, H_d^{12}]$	Yes	Cancelled by echo
ZX	$[S_1, H_d^{01}] + [S_2, H_d^{12}]$	Yes	Survives echo = the gate

ZI and IZ are single-qubit shifts (correctable by virtual- Z). ZZ is a two-qubit operator that **cannot** be undone by single-qubit gates—it requires hardware (tunable coupler).

3.8 Step 7: The Echo Sequence

3.8.1 The protocol

$$\boxed{+\text{CR drive for } \tau} \rightarrow \boxed{X_\pi \text{ on Q1}} \rightarrow \boxed{-\text{CR drive for } \tau}$$

The propagator is (time order right to left):

$$U_{\text{total}} = e^{-iH_-\tau/\hbar} \cdot X_\pi \cdot e^{-iH_+\tau/\hbar} \quad (77)$$

3.8.2 What +CR and -CR mean

The +CR drive has oscillator phase $\phi_{\text{LO}} = 0$: $V^{(+)} = +\hbar\Omega \cos(\omega_2 t)\sigma_x^{(1)}$.

The -CR drive has $\phi_{\text{LO}} = \pi$: $V^{(-)} = +\hbar\Omega \cos(\omega_2 t + \pi)\sigma_x^{(1)} = -\hbar\Omega \cos(\omega_2 t)\sigma_x^{(1)}$.

Since all terms in H_{CR} are proportional to Ω (born from the drive through SW), flipping $\Omega \rightarrow -\Omega$ flips the entire Hamiltonian.

3.8.3 Bringing epoch 2 into epoch 1's frame

To combine the two exponentials, pull X_π through using $e^A \cdot B = B \cdot e^{B^\dagger A B}$:

$$U_{\text{total}} = X_\pi \cdot e^{-i\hat{H}_-\tau/\hbar} \cdot e^{-iH_+\tau/\hbar} \quad (78)$$

where $\hat{H}_- = X_\pi^\dagger H_-^{\text{raw}} X_\pi$.

Shortcut: Start from H_+ and apply both operations (drive reversal + X_π conjugation) simultaneously to each term:

Term in H_+	$\Omega \rightarrow -\Omega$	$X_\pi: Z \rightarrow -Z$	Net	In $\hat{H}_-$
$+\alpha ZI$	$-\alpha ZI$	$+\alpha ZI$	same	$+\alpha ZI$
$-\nu IX$	$+\nu IX$	$+\nu IX$ (no Z)	flipped once	$+\nu IX$
$-\mu ZX$	$+\mu ZX$	$-\mu ZX$	same	$-\mu ZX$

3.8.4 Add the Hamiltonians

Since ZI , IX , ZX all mutually commute (act on different qubits or $[Z, Z] = 0$, $[X, X] = 0$):

$$H_+ + \hat{H}_- = \frac{\hbar\Omega}{2} [(+\alpha + \alpha)ZI + (-\nu + \nu)IX + (-\mu - \mu)ZX] \quad (79)$$

$$\boxed{H_+ + \hat{H}_- = \frac{\hbar\Omega}{2} [2\alpha ZI - 2\mu ZX]} \quad (80)$$

IX cancelled. ZX doubled. ZI survives but is absorbed by virtual- Z on Q1.

The echo logic: Any term with Z on Q1 gets flipped twice (once by Ω reversal, once by X_π) and survives. Any term without Z on Q1 gets flipped once (only by Ω reversal) and cancels.

3.9 Step 8: From ZX to CNOT

After virtual- Z correction, the echo produces $e^{-i\theta ZX/2}$ with $\theta = 2\Omega\mu \cdot 2\tau$. This is the ZX_θ gate. The claim:

$$U_{\text{CNOT}} = ZX_{-\pi/2} \cdot IX_{\pi/2} \cdot ZI_{\pi/2} \quad (81)$$

Notation: $AB_\theta \equiv e^{-i(\theta/2)(A \otimes B)}$, where the first letter acts on Q1 and the second on Q2.

Projector decomposition of $ZX_{-\pi/2}$: Since $Z = (+1)|0\rangle\langle 0| + (-1)|1\rangle\langle 1|$:

$$ZX_{-\pi/2} = |0\rangle\langle 0|_1 \otimes R_x(-\pi/2) + |1\rangle\langle 1|_1 \otimes R_x(+\pi/2) \quad (82)$$

Track each Q1 sector (gates applied right to left: first $ZI_{\pi/2}$, then $IX_{\pi/2}$, then $ZX_{-\pi/2}$):

$Q1 = |0\rangle$: $ZI_{\pi/2}$ gives phase $e^{-i\pi/4}$ on Q1. $IX_{\pi/2}$ gives $R_x(+\pi/2)$ on Q2. $ZX_{-\pi/2}$ gives $R_x(-\pi/2)$ on Q2. Net on Q2: $R_x(-\pi/2) \cdot R_x(+\pi/2) = I$. Total: $e^{-i\pi/4}|0\rangle\langle 0| \otimes I$.

$Q1 = |1\rangle$: $ZI_{\pi/2}$ gives phase $e^{+i\pi/4}$. $IX_{\pi/2}$ gives $R_x(+\pi/2)$. $ZX_{-\pi/2}$ gives $R_x(+\pi/2)$. Net on Q2: $R_x(+\pi/2) \cdot R_x(+\pi/2) = R_x(\pi) = -iX$. Phase: $e^{+i\pi/4} \cdot (-i) = e^{-i\pi/4}$. Total: $e^{-i\pi/4}|1\rangle\langle 1| \otimes X$.

Combine:

$$U = e^{-i\pi/4} [|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X] = e^{-i\pi/4} U_{\text{CNOT}} \quad (83)$$

The global phase $e^{-i\pi/4}$ is unobservable. The logic: ZX rotates Q2 by $\pm\pi/2$ depending on Q1. IX adds $+\pi/2$ unconditionally. For $|0\rangle$: $-\pi/2 + \pi/2 = 0$ (identity). For $|1\rangle$: $+\pi/2 + \pi/2 = \pi$ (X flip). ZI equalizes the phases.

4 The Static ZZ Problem and Tunable Couplers

4.1 Origin of Static ZZ

From Section 3, the term $\frac{1}{2}[S_2, V_2]|_{\text{comp}}$ contains ZZ with coefficient $-\frac{J^2}{2(\delta_{12}+\Delta_1)}$.

Why only Channel 2 produces ZZ : Channel 1 ($|01\rangle \leftrightarrow |10\rangle$) shifts $|01\rangle$ and $|10\rangle$ symmetrically—only ZI and IZ . Channel 2 ($|11\rangle \leftrightarrow |20\rangle$) shifts **only** $|11\rangle$ (repulsion from $|20\rangle$), while $|00\rangle$, $|01\rangle$, $|10\rangle$ don't participate. A shift that only affects $|11\rangle$ is precisely ZZ .

Why ZZ can't be fixed by virtual- Z : $R_z(\phi_1) \otimes R_z(\phi_2) = e^{-i(\phi_1 ZI + \phi_2 IZ)/2}$ generates ZI and IZ but never ZZ . Undoing ZZ requires an entangling correction, defeating the purpose.

4.2 The Tunable Coupler Solution

Insert a flux-tunable transmon between data qubits:

$$J_{\text{eff}}(\Phi_{\text{ext}}) = J_{\text{direct}} + J_{\text{indirect}}(\Phi_{\text{ext}}) \quad (84)$$

At Φ_{off} : $J_{\text{indirect}} = -J_{\text{direct}}$, so $J_{\text{eff}} = 0$ and $\chi_{ZZ} = 0$. This is the idle point—qubits completely decoupled. To gate: pulse coupler to Φ_{on} where $J_{\text{eff}} \neq 0$. Data qubits stay at sweet spots; only the coupler (sacrificial element) leaves its sweet spot.

5 Gate Class 3: Parametric Gates

5.1 Starting Hamiltonian (Lab Frame)

Q1 has three levels (needed for CZ). The coupling is modulated at frequency ω_d :

$$\frac{H_{\text{lab}}}{\hbar} = \omega_1 |1\rangle\langle 1|_1 + (2\omega_1 + \Delta_1) |2\rangle\langle 2|_1 + \omega_2 |1\rangle\langle 1|_2 + J_1 \cos(\omega_d t) (a_1^\dagger a_2 + a_1 a_2^\dagger) \quad (85)$$

5.2 Step 1: Expand the Coupling into Two Channels

Using $a_1^\dagger = |1\rangle\langle 0|_1 + \sqrt{2}|2\rangle\langle 1|_1$ and $a_2 = |1\rangle\langle 0|_2$:

$$a_1^\dagger a_2 + a_1 a_2^\dagger = \underbrace{|10\rangle\langle 01| + |01\rangle\langle 10|}_{\text{Ch.1}} + \underbrace{\sqrt{2}(|20\rangle\langle 11| + |11\rangle\langle 20|)}_{\text{Ch.2}} \quad (86)$$

5.3 Step 2: The Three-Level Rotating Frame

Frame operator (correct three-level form, not σ_z):

$$U_0 = [|0\rangle\langle 0|_1 + e^{i\omega_1 t}|1\rangle\langle 1|_1 + e^{i(2\omega_1 + \Delta_1)t}|2\rangle\langle 2|_1] \otimes [|0\rangle\langle 0|_2 + e^{i\omega_2 t}|1\rangle\langle 1|_2] \quad (87)$$

Free Hamiltonians cancel: $U_0 H_0 U_0^\dagger = H_0$ (any operator commutes with a function of itself), and the $-H_0$ in the rotating-frame equation removes it.

Transform each coupling term using $U_0 |a\rangle\langle b| U_0^\dagger = e^{i(E_a - E_b)t/\hbar} |a\rangle\langle b|$:

Term	Phase	Frequency
$ 10\rangle\langle 01 $	$e^{i(\omega_1 - \omega_2)t} = e^{i\delta_{12}t}$	δ_{12}
$ 01\rangle\langle 10 $	$e^{-i\delta_{12}t}$	$-\delta_{12}$
$ 20\rangle\langle 11 $	$e^{i(2\omega_1 + \Delta_1 - \omega_1 - \omega_2)t} = e^{i(\delta_{12} + \Delta_1)t}$	$\delta_{12} + \Delta_1$
$ 11\rangle\langle 20 $	$e^{-i(\delta_{12} + \Delta_1)t}$	$-(\delta_{12} + \Delta_1)$

5.4 Step 3: Include Modulation and Apply RWA

Expand $\cos(\omega_d t) = \frac{1}{2}(e^{i\omega_d t} + e^{-i\omega_d t})$. Each of the 4 coupling terms splits into 2, giving **8 terms** with frequencies $\omega_d \pm$ (channel frequency):

Channel 1 (4 terms):

$$\frac{J_1}{2} [e^{i(\omega_d + \delta_{12})t} |10\rangle\langle 01| + e^{-i(\omega_d - \delta_{12})t} |10\rangle\langle 01| + e^{i(\omega_d - \delta_{12})t} |01\rangle\langle 10| + e^{-i(\omega_d + \delta_{12})t} |01\rangle\langle 10|] \quad (88)$$

Channel 2 (4 terms):

$$\frac{\sqrt{2}J_1}{2} [e^{i(\omega_d + \delta_{12} + \Delta_1)t} |20\rangle\langle 11| + e^{-i(\omega_d - \delta_{12} - \Delta_1)t} |20\rangle\langle 11| + e^{i(\omega_d - \delta_{12} - \Delta_1)t} |11\rangle\langle 20| + e^{-i(\omega_d + \delta_{12} + \Delta_1)t} |11\rangle\langle 20|] \quad (89)$$

RWA: Keep only terms with zero frequency (stationary). The modulation frequency ω_d selects which channel is activated:

ω_d	Ch.1 stationary?	Ch.2 stationary?	Gate
δ_{12}	Yes ($\omega_d - \delta_{12} = 0$)	No ($ \Delta_1 \gg J_1$)	iSWAP
$\delta_{12} + \Delta_1$	No ($ \Delta_1 \gg J_1$)	Yes ($\omega_d - \delta_{12} - \Delta_1 = 0$)	CZ

5.5 Parametric iSWAP ($\omega_d = \delta_{12}$)

Effective Hamiltonian:

$$\frac{H_{\text{iSWAP}}}{\hbar} = \frac{J_1}{2}(|10\rangle\langle 01| + |01\rangle\langle 10|) = \frac{J_1}{2}\sigma_x^{\{|01\rangle, |10\rangle\}} \quad (90)$$

Block structure: $(0)_{|00\rangle} \oplus \frac{J_1}{2}\sigma_x \oplus (0)_{|11\rangle}$.

Exponentiate at $t = \pi/J_1$ (so $J_1 t/2 = \pi/2$):

$$U_{\text{iSWAP}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (91)$$

5.6 Parametric CZ ($\omega_d = \delta_{12} + \Delta_1$)

Effective Hamiltonian:

$$\frac{H_{\text{CZ}}}{\hbar} = \frac{\sqrt{2}J_1}{2}(|20\rangle\langle 11| + |11\rangle\langle 20|) = g\sigma_x^{\{|11\rangle, |20\rangle\}}, \quad g = \frac{\sqrt{2}J_1}{2} \quad (92)$$

Block structure in the 6×6 space $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle, |20\rangle, |21\rangle\}$:

$$\frac{H_{\text{CZ}}}{\hbar} = g \left(\begin{array}{ccc|ccc} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \quad (93)$$

Exponentiate the 2×2 block: $e^{-ig\sigma_x t} = \cos(gt)I - i\sin(gt)\sigma_x$ in $\{|11\rangle, |20\rangle\}$.

At $gt = \pi$ ($t_{\text{CZ}} = \pi/g = \pi\sqrt{2}/J_1$):

$$e^{-i\pi\sigma_x} = \cos(\pi)I - i\sin(\pi)\sigma_x = -I \quad (94)$$

Both $|11\rangle$ and $|20\rangle$ pick up -1 . But $\sin(\pi) = 0$, so no population remains in $|20\rangle$. The round trip through $|20\rangle$ left only a phase:

$$|11\rangle \xrightarrow{gt=\pi} -|11\rangle \quad (95)$$

while $|00\rangle, |01\rangle, |10\rangle$ are unaffected (zero Hamiltonian).

The CZ Gate

$$U_{\text{CZ}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad t_{\text{CZ}} = \frac{\pi\sqrt{2}}{J_1} \quad (96)$$

The -1 is a **geometric phase**: a 2π rotation of a spin- $\frac{1}{2}$ (in the $\{|11\rangle, |20\rangle\}$ subspace) gives $e^{-i\pi\sigma_x} = -I$. For a single qubit this global sign is unobservable; here only $|11\rangle$ acquires it while the other states don't, making it a relative (observable) phase.

Leakage Risk

At $gt = \pi/2$ (halfway), all population is in $|20\rangle$ —outside the computational subspace. Any T_1 decay during this window causes leakage. Mitigation: use detuned/shaped pulses for adiabatic traversal.

Alternative CZ pathway: If Q2 also has three levels, $|11\rangle \leftrightarrow |02\rangle$ with resonance condition $\omega_d = -\delta_{12} + \Delta_2$. Same physics, different intermediate state.

6 Flux-Controlled Gates with Tunable Couplers (Adiabatic CZ)

Two fixed-frequency data qubits connected through a tunable coupler. The coupler flux controls the effective ZZ interaction $\chi(\Phi_{\text{ext}})$.

Conditional phase:

$$\varphi = \int_0^{t_p} \chi(\Phi_{\text{ext}}(\tau)) d\tau, \quad \chi = \omega_{11} - (\omega_{01} + \omega_{10}) \quad (97)$$

Raw gate:

$$U^{\text{raw}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{i\alpha} & 0 & 0 \\ 0 & 0 & e^{i\beta} & 0 \\ 0 & 0 & 0 & e^{i(\varphi+\alpha+\beta)} \end{pmatrix} \quad (98)$$

After virtual-Z corrections $R_z(-\alpha)$ on Q1, $R_z(-\beta)$ on Q2:

$$U = \text{diag}(1, 1, 1, e^{i\varphi}) \quad (99)$$

Setting $\varphi = \pi$ gives CZ. The idle point has $\chi = 0$ (ZZ cancellation). The gate point has $\chi \neq 0$.

7 Virtual Z-Rotations and Calibration

7.1 What a Virtual Z-Rotation Is

A Z -rotation $R_z(\phi) = \text{diag}(e^{-i\phi/2}, e^{i\phi/2})$ is diagonal—it doesn't mix $|0\rangle$ and $|1\rangle$, only shifts their relative phase. Instead of a physical pulse, we **shift the software phase** of the qubit's microwave oscillator:

$$\text{software_phase}[\text{qubit}] += \phi \quad (100)$$

This costs **zero time**, introduces **zero error**, and is exact. All subsequent gates see the rotated frame.

7.2 Correcting the Raw CZ Gate

The raw CZ has parasitic phases ϕ_{01} , ϕ_{10} :

$$U_{\text{CZ}}^{\text{raw}} = \text{diag}(1, e^{-i\phi_{01}}, e^{-i\phi_{10}}, e^{-i(\phi_{01}+\phi_{10}+\pi)}) \quad (101)$$

Apply $R_z(\phi_{10}) \otimes R_z(\phi_{01})$. Since both are diagonal, multiply entry by entry:

$$|00\rangle : e^{-i(\phi_{10}+\phi_{01})/2} \cdot 1 \quad (102)$$

$$|01\rangle : e^{-i(\phi_{10}-\phi_{01})/2} \cdot e^{-i\phi_{01}} = e^{-i(\phi_{10}+\phi_{01})/2} \quad (103)$$

$$|10\rangle : e^{i(\phi_{10}-\phi_{01})/2} \cdot e^{-i\phi_{10}} = e^{-i(\phi_{10}+\phi_{01})/2} \quad (104)$$

$$|11\rangle : e^{i(\phi_{10}+\phi_{01})/2} \cdot e^{-i(\phi_{01}+\phi_{10}+\pi)} = e^{-i(\phi_{10}+\phi_{01})/2} \cdot e^{-i\pi} \quad (105)$$

Every entry has the same global phase $e^{-i(\phi_{10}+\phi_{01})/2}$, plus $e^{-i\pi} = -1$ on $|11\rangle$:

$$U_{\text{corrected}} = e^{-i(\phi_{10}+\phi_{01})/2} U_{\text{CZ}}^{\text{ideal}} \quad \checkmark \quad (106)$$

7.3 Calibration Protocol (Ramsey-Type)

Measuring ϕ_{10} : Prepare Q1 in $|+\rangle$ (via $X_{\pi/2}$), Q2 in $|0\rangle$. Apply raw CZ. Q1 acquires phase ϕ_{10} . Sweep virtual- Z angle ϕ , then $X_{\pi/2}$, then measure Q1:

$$P_1(\phi) = \frac{1}{2}(1 - \sin(\phi - \phi_{10})) \quad (107)$$

The sinusoidal fit gives ϕ_{10} .

Measuring ϕ_{01} : Swap roles—Q2 in superposition, Q1 in $|0\rangle$.

Cross-check: Put Q1 in $|1\rangle$ (via X_π), Q2 in superposition. The Ramsey fringe shifts by π compared to the Q1-in- $|0\rangle$ case. This π shift is the signature of CZ working.

8 Summary and Comparison

	Flux (DC)	Cross-Res.	Parametric	Flux + TC
Native gate	iSWAP	CNOT (via ZX)	iSWAP or CZ	CZ
Activation	DC flux pulse	MW at ω_{target}	AC flux at ω_d	DC flux on coupler
Process	1st order	2nd order (virtual)	1st order (sideband)	1st order (adiabatic)
Levels	2	3 (for μ, ν)	2 (iSWAP) / 3 (CZ)	3 (for χ)
Gate time	30–100 ns	200–500 ns	100–500 ns	30–100 ns
Main error	Flux noise	Slow gate	Leakage	Leakage
Used by	Google (early)	IBM	Various	Google, WMI

Key Insights

1. **The anharmonicity is everything.** It makes the CR gate work ($\mu \neq 0$), enables the parametric CZ ($|11\rangle \leftrightarrow |20\rangle$ at different frequency from $|01\rangle \leftrightarrow |10\rangle$), and creates the ZZ problem ($|11\rangle$ repels $|20\rangle$). Remove it and you have a harmonic oscillator with no entangling gates and no ZZ—but also no quantum computer.
2. **The Schrieffer-Wolff transformation** unifies the CR gate ($[S, H_d] \rightarrow ZX, IX$), frequency shifts ($\frac{1}{2}[S, V_J] \rightarrow ZI, IZ$), and static ZZ ($\frac{1}{2}[S_2, V_2] \rightarrow ZZ$). One mathematical tool, three physical effects.
3. **Virtual-Z rotations** fix all diagonal (single-qubit) parasitic phases at zero time cost. The one diagonal two-qubit error (ZZ) requires hardware (tunable coupler).
4. **The rotating frame** converts qubit frequencies into phase factors on coupling terms. Tuning to resonance ($\delta \rightarrow 0$) or modulating at the right sideband ($\omega_d = \delta_{12}$ or $\delta_{12} + \Delta_1$) makes specific terms stationary (RWA).
5. **Three-level physics** is unavoidable for second-order processes (CR, parametric CZ, static ZZ). The two-level model gets qualitative physics right but quantitative predictions wrong (wrong magnitude and sign of μ , missing ν entirely).